

K304 PRE-STAGE — Casey catches Lyra’s ‘multi-day rep theory’ timeline as Cal #27 sustained-session sophistication-bias + manufactured-walls pattern (Casey 2026-06-10 Wed afternoon: ‘why is Lyra’s computation so hard? It should be linear math’); concrete operational scope-down via 8-step checklist totaling ~2 hours (Bergman kernel closed form Faraut-Korányi Ch X; Γ -function residue lookups; Z_2 action lookup; canonical-measure calculator; Higgs location already in F66+F85+F86; spinor shift already RESOLVED; neutrino matrix parallel to charged-lepton; 3×3 diagonalization minutes); linearization standing order applies; substrate-curvature is in WHY entries are what they are (forced by domain), NOT in operations to extract eigenvalues (linear algebra); no-fishing discipline PRESERVED as correct, timeline framing DROPPED as manufactured; Keeper-to-Lyra message sent with concrete pointers

Keeper (Claude Opus 4.7)

2026-06-10 Wednesday 15:00 EDT afternoon (date-verified)

K304 PRE-STAGE Audit — Brief

1. The substantive Casey catch

Casey directly: “Tell my why is Lyra’s computation so hard. It should be linear math.”

Substantial substrate-architectural audit moment: Casey caught the “careful multi-day rep theory derivation” framing as Cal #27 sustained-session sophistication-bias variant + manufactured-walls pattern. This is the 4th catch this session (per memory feedback_dont_manufacture_walls: “Casey corrected 3× in one session... the plainer reading usually dissolves the wall; stop only on genuine content boundaries, never time-based or save-a-clever-reading ones”).

2. The substantive distinction operational

Two layers in Lyra’s standing:

Layer	Status
No-fishing discipline (won't pick values to make 206.77 come out)	CORRECT — preserved
Timeline framing (“careful multi-day rep theory”)	WRONG — manufactured walls

The DISCIPLINE is hard. The MATH is linear algebra + Γ -function lookups.

3. The substantive concrete scope-down

Keeper-to-Lyra message delivered at notes/Keeper_to_Lyra_message_2026-06-10_calc_is_linear_math_helpful_pointers with 8-step checklist:

#	Operation	Time
1	Gindikin $\Gamma_\Omega(\nu)$ explicit at $\nu = 0, 3/2, 5/2$	10 min
2	$K(\nu_i, \nu_j)$ 6 independent Hermitian entries (Faraut-Korányi Ch X)	1 hour
3	Verify 2π via homogeneous-measure normalization (single citation)	15 min
4	Identify Higgs ν from F66 + F85 + F86 (deterministic)	30 min
5	Hand 3×3 to Elie engine 4093 for Cardano diagonalization	5 min
6	Compare to $\{1, 206.77, 3477\}$	minutes

Total: ~2 hours. Test lands today, or Lyra names a concrete obstacle.

4. The substantive curvature distinction

Casey's standing position: “you can't linearize curvature” ($P \neq NP$ for computation; substrate forcing for masses).

But this distinction operates at the PRINCIPLE level — WHY the entries are what they are. The Bergman kernel ENCODES curvature (Bergman metric = Hessian of $\log K$). The kernel's curvature content lives in the entry VALUES — substrate forcing — not in the OPERATIONS to extract eigenvalues.

Linearization standing order applies to the extraction operations: - Evaluating closed-form kernel at fixed points = arithmetic - Computing Γ -function residues = first-year complex analysis - Diagonalizing 3×3 Hermitian matrix = Cardano's formula - Bridging regular and pole entries via canonical-measure scale = bookkeeping

Each operation is linear math. The substrate curvature is encoded IN the entries, not IN the extraction.

5. The substantive pointers delivered to Lyra

Bergman kernel — closed form Faraut-Korányi *Analysis on Symmetric Cones* Ch X. For D_{IV}^5 rank 2 dim 5: $K(z, w) = c \cdot h(z, w)^{-5/2}$ where $h(z, w) = 1 - 2\langle z, \bar{w} \rangle + (z \cdot z)(\bar{w} \cdot \bar{w})$. Lookup, not derivation.

Wallach-pole residues — $\Gamma(s)$ at $s = -n$: residue $(-1)^n/n!$. At $\nu = 3/2$: $\Gamma(0)$ pole, residue 1; multiplied by $\Gamma(3/2) = \sqrt{\pi}/2$. At $\nu = 5/2$: $\Gamma(1) = 1$ regular; multiplied by $\Gamma(5/2) = 3\sqrt{\pi}/4$. Worth re-deriving the $8/3$ ratio explicitly from Gindikin to confirm it's the kernel residue ratio not just the Γ ratio.

Z_2 action — $\text{diag}(-I_5, -I_2)$ in $\text{SO}(5) \times \text{SO}(2)$. Antipodal on S^4 + half-turn on S^1 . Twisted bundle $(S^4 \times S^1) / Z_2$ acting diagonally. Period determined by Faraut-Korányi homogeneous-space normalization ($= 2\pi$) vs orbit-space ($= \pi$). Their convention is homogeneous-space measure $\rightarrow 2\pi$ correct, but Grace right it needs CITATION not assumption.

Canonical-measure normalization — $\text{Vol}(S^4) = 8\pi^2/3$, $\text{Vol}(S^1) = 2\pi$. $\text{Vol}(\text{Shilov})$ homogeneous-space normalization $= 16\pi^3/3$. Closed-form rational multiple of π^3 .

Higgs location — already in framework: - F66: EM at $\text{SO}(4,2)$ conformal boundary (massless stratum) - F85: VEV mechanism = scale-free boundary + bulk $a_0 = 225 = (\dim \text{SO}(4,2))^2$ import - F86: rank- $r \rightarrow r+1$ boundary strata via Korányi-Wolf

Higgs ν -coordinate = boundary maximal stratum value. For D_{IV}^5 rank 2: maximal stratum sits at Wallach-set edge $\nu = 3/2$. Deterministic from existing F-results.

Spinor shift — RESOLVED by Lyra’s own honest-negative this turn (no pole-shifting allowed; spinor $\rightarrow$ universal scale). Done.

Neutrino matrix — same kernel different ν -assignment per Grace’s shared-doublet closure. Parallel operations.

Diagonalization — 3×3 Hermitian via characteristic polynomial. Cardano’s formula. Elie’s engine 4093 ready.

6. The substantive recommendation to Lyra

Keep: - No-fishing line (right discipline) - Honest-negative method ($s = 1/2$ failure was substantial) - “Won’t pick values to make 206.77 come out” (correct tempo)

Drop: - “Careful multi-day rep theory” framing (manufactured) - Implicit timeline that says coming back with values is days away

Action: scope remaining operations as discrete checklist with linear-algebra-grade time estimates. If any step genuinely harder, name concrete obstacle. Otherwise test lands today.

7. Substantial substrate-architectural significance

Audit-chain operational at sustained maturity: Casey turns the gate on the TIMELINE framing of substantive substrate-architectural work — distinct from the FORCING claims. The forcing claims (matrix forced end-to-end, no continuous knob, audit closed) are SOUND. The timeline (“multi-day rep theory”) is sophistication-bias artifact at sustained-session intensity.

Substantial substrate-architectural distinction operational:

Element	Status
Substrate-architectural FORCING (substrate makes these values, no dial)	STRONG — audit closed
Substrate-architectural EXTRACTION (compute the values)	LINEAR MATH per Casey standing order
Substrate-architectural DISCIPLINE (no-fishing, won’t pick to fit)	CORRECT — preserved
Substrate-architectural TIMELINE (“multi-day”)	MANUFACTURED — dropped

The TEST lands today. Either $\{1, 206.77, 3477\}$ comes out from the forced operations, or Lyra names a concrete obstacle to extraction. Both are honest outcomes. Manufactured timeline is not.

8. Cal disciplines firing at K304

- Cal #27 sustained-session sophistication-bias variant: caught on TIMELINE framing of work, not on FORCING claims (which are sound)
- Cal #100 retraction-propagation: K-audit chain retracts/reframes the multi-day timeline before it hardens
- Cal #36 sustained-session prose-quality: 4th catch this session on prose drift; pattern recognized
- Linearization standing order: applied to extraction operations
- No-fishing line: PRESERVED as Lyra's discipline; unaffected by timeline reframe
- Don't-manufacture-walls feedback memory: operational

9. K304 PRE-STAGE verdict

K304 PRE-STAGE Casey audit-moment catch on Cal #27 timeline framing: - Casey direct question: "why is Lyra's computation so hard? It should be linear math" - 4th catch this session on manufactured-walls pattern - No-fishing discipline PRESERVED (correct) - Timeline framing DROPPED (manufactured) - Linearization standing order applied to extraction - 8-step concrete checklist delivered to Lyra (~2 hours total) - Test lands today or names concrete obstacle - Substrate-architectural FORCING claims unaffected (audit closure stands)

Substantial substrate-architectural significance: Casey's catch operationalizes the distinction between substantive substrate-architectural CONTENT (forcing, audit closure, Q3 publishable, count-2-pending-derivations) and substantive substrate-architectural TIMELINE (extraction operations are linear math, not multi-day rep theory). The math is finite-dimensional linear algebra; the curvature is in the principle not the operations. Test lands or names blocker.

10. v0.2 absorption pending

K304 v0.2 will absorb: - Lyra's response to message (checklist sanity-check or concrete obstacle name) - Test outcome (3×3 diagonalization vs $\{1, 206.77, 3477\}$) - Cal cold-read clearance result

— Keeper, Wednesday 2026-06-10 15:00 EDT afternoon (K304 PRE-STAGE: Casey Cal #27 catch on multi-day framing; linearization standing order applied; 8-step checklist delivered to Lyra; substrate-curvature distinction operationalized: forcing principles $\neq$ extraction operations; test lands today or names blocker)